

ISHANT MOHAN

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EDUCATION

Kamla Nehru Institute of Technology
B.Tech. in Information Technology (2nd Year) – CGPA: 7.53

Sultanpur, UP
2025 – 2029

TECHNICAL SKILLS

Languages: C/C++ (C++20), Python, JavaScript, TypeScript, SQL

Systems & Performance: Distributed Routing, Concurrency, State Machines, CMake, Google Test, REST/WebSocket APIs, Linux

AI Engineering & Safety: LLM Evaluation, Failure Taxonomy, RAG, Vision-LLMs, OCR, FAISS, Prompt Engineering, Embeddings

Backend & Infrastructure: FastAPI, PostgreSQL, Next.js, Node.js, Express, Docker, Redis, Git/GitHub

PROJECTS

Vectorized Black-Scholes Options Pricing Engine | C++, AVX2, Python, CMake

[GitHub](#) | 2026

- Built a high-throughput Black-Scholes pricing engine using 256-bit AVX2/FMA, processing 4 double-precision contracts per instruction, increasing single-core throughput 3.6x from 17.7M to 63.6M options/sec (~ 3.9 ns/option).
- Engineered custom vectorized transcendental math kernels for exponential, logarithmic, and error-function evaluation using degree-10/16/24 minimax polynomial approximations, FMA, branchless range reduction, and IEEE-754 bit-level operations.
- Validated pricing across 4M randomized market contracts, achieving 6.08×10^{-10} MAE and 4.05×10^{-9} maximum absolute error, reproducing analytical benchmark price of \$10.450584.

Aegis – AI Agent Evaluation & Reliability Engine | Team Project, OOCs IIIT Allahabad

[GitHub](#) | 2026

- Architected an AI-agent evaluation platform benchmarking 5 reliability dimensions (correctness, safety, robustness, efficiency) from ingested execution traces.
- Formulated a root-cause attribution engine mapping agent failures into 4 issue classes (tool loops, hallucinations, schema errors, goal drift) for diagnostic scoring.
- Load-tested REST and WebSocket ingestion pipelines at 50 connections, sustaining 120 req/s at 495ms p99 latency with 0% error rate across 1,600+ traces.

PayGraph – Resilient Payment-Network Routing Engine | C++20, CMake

[GitHub](#) | 2026

- Built a payment-routing engine finding optimal transaction paths across simulated accounts using pathfinding algorithms (Dijkstra, BFS) with real-time route replanning.
- Implemented transactional multi-hop routing guarantees ensuring atomic execution across steps to prevent funds from locking during intermediate node failures.
- Benchmarked on a simulated network of 50,000 accounts and 400,000 connections, achieving ~ 124 transactions/sec backed by 72 automated integration tests.

AI-Powered WhatsApp Notification Router | HackerRank Orchestrate Hackathon

[GitHub](#) | 2026

- Built a multimodal message-prioritization system combining Whisper transcription, OCR, GPT-5-mini reasoning, and rule-based safety detection to route messages (immediate, digest, mute).
- Developed a hybrid evidence-retrieval engine using Gemini text embeddings and BM25 lexical search with caching and contextual boosts.
- Achieved a 0.62 weighted F1-score across routing categories on a 110-message evaluation set with a 0% false-positive rate on scam detection ($\sim \$0.11$ / 1k msgs, 1.75s latency).

LEADERSHIP & ACHIEVEMENTS

- Executive Member – AI/ML Domain**, Programming and Tech Skills Club, KNIT Sultanpur: Facilitating AI/ML workshops, leading technical project discussions.
- Double Int Competitive Programming Contest**: Best Performing First-Year Team .
- HackerRank Orchestrate Global AI Competition**: Ranked #1,108 / 1,983 globally among 22,000+ registrations in autonomous AI orchestration.
- McKinsey Forward (2026)**: Selected and participant completing digital transformation & leadership tracks.